Lecture Scribe

Course: CSE 400 – Fundamentals of Probability in Computing
Topic: Core Probability Models and Techniques Required for Tutorial-2
Institution: School of Engineering and Applied Science (SEAS), Ahmedabad University

This lecture scribe reconstructs the probability concepts, distributions, and reasoning techniques required to solve all problems appearing in Tutorial-2. The document is organized conceptually rather than question-wise. It includes definitions, assumptions, formulas, derivation ideas, and reasoning steps necessary to solve problems involving uniform distributions, normal distributions and likelihood comparisons, expectation minimization, exponential distributions, conditional probabilities, Poisson variables, transformations of random variables, approximation methods, and order statistics.

All material presented here is directly derived from the concepts required by the problems in Tutorial-2.

1. Random Variables and Probability Models

A **random variable** is a function that assigns a numerical value to each outcome of a random experiment.

Two primary types are used in this tutorial:

Continuous Random Variables

A random variable X is continuous if it has a **probability density function (pdf)** $f_X(x)$ such that

$$P(a \le X \le b) = \int_a^b f_X(x)\,dx$$

and

$$f_X(x) \ge 0, \qquad \int_{-\infty}^{\infty} f_X(x)dx = 1$$

The **cumulative distribution function (CDF)** is

$$F_X(x) = P(X \le x) = \int_{-\infty}^{x} f_X(t)\,dt$$

The CDF fully determines the distribution.

Discrete Random Variables

A discrete random variable has a **probability mass function (pmf)**

$$P(X = x)$$

with

$$\sum_x P(X = x) = 1$$

2. Uniform Distribution on an Interval

A **uniform distribution** models a situation where every point in an interval is equally likely.

If

$$X \sim \text{Uniform}(a, b)$$

then

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \le x \le b \\ 0, & \text{otherwise} \end{cases}$$

and

$$F_X(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \le x \le b \\ 1, & x > b \end{cases}$$

Interpretation of "choosing a point at random on a line segment"

This means the position X of the chosen point is modeled as

$$X \sim \text{Uniform}(0, L)$$

Thus the probability of the point falling in any subinterval is proportional to its length.

If a point divides a segment into two pieces X and $(L - X)$, the analysis of ratios or inequalities between these quantities is performed by expressing the event in terms of the random variable X.

3. Normal Distribution and Likelihood Comparison

The **normal distribution** is defined by mean μ and variance σ^2.

If

$$X \sim N(\mu, \sigma^2)$$

then the density is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Likelihood-Based Classification

When an observation could come from two possible normal distributions, classification is based on **likelihood comparison**.

Suppose an observation x may come from two populations:

$$f_1(x) \quad \text{and} \quad f_2(x)$$

If prior probabilities are

$$P(\text{population 1}) = p_1, \qquad P(\text{population 2}) = p_2$$

then the posterior comparison uses

$$p_1 f_1(x) \quad \text{vs} \quad p_2 f_2(x)$$

This follows from **Bayes' theorem**:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Error probabilities are analyzed by equating the probabilities of misclassification under the two hypotheses.

4. Minimizing Expected Absolute Deviation

A common optimization problem in probability is minimizing

$$E[|X-a|]$$

for some constant a.

Key Result

The value of a that minimizes $E[|X-a|]$ is the **median of the distribution of X**.

Reasoning

1. Write

$$E[|X-a|] = \int |x-a| f_X(x)dx$$

2. Split the integral at $x = a$

$$E[|X-a|] = \int_{-\infty}^{a} (a-x) f_X(x) dx + \int_{a}^{\infty} (x-a) f_X(x) dx$$

3. Differentiate with respect to a
4. Set derivative equal to zero

This leads to

$$P(X \le a) = P(X \ge a)$$

which defines the **median**.

Uniform Case

For

$$X \sim \text{Uniform}(0, A)$$

the median equals

$$A/2$$

Exponential Case

For

$$X \sim \text{Exponential}(\lambda)$$

the median satisfies

$$F_X(a) = \frac{1}{2}$$

where

$$F_X(x) = 1 - e^{-\lambda x}$$

5. Exponential Distribution

The exponential distribution models **waiting times between events**.

If

$$X \sim \text{Exponential}(\lambda)$$

then

$$f_X(x) = \lambda e^{-\lambda x}, \qquad x \ge 0$$

$$F_X(x) = 1 - e^{-\lambda x}$$

Mean and Variance

$$E[X] = \frac{1}{\lambda}$$

$$Var(X) = \frac{1}{\lambda^2}$$

Memoryless Property

The exponential distribution satisfies

$$P(X > s + t \mid X > t) = P(X > s)$$

This means that the remaining lifetime does not depend on how long the system has already survived.

This property is fundamental when analyzing remaining lifetimes or additional operating time.

6. Conditional Probability with Mixtures of Exponential Variables

Sometimes a system may belong to one of several types with different exponential parameters.

Suppose:

Type i battery has lifetime

$$X_i \sim \text{Exponential}(\lambda_i)$$

The probability of selecting type i is

$$p_i$$

Then the lifetime distribution of a randomly selected battery is a **mixture distribution**.

Survival Probability

$$P(X > t) = \sum_i p_i e^{-\lambda_i t}$$

Conditional Probability of Additional Lifetime

Using conditional probability,

$$P(X > t + s \mid X > t) = \frac{P(X > t + s)}{P(X > t)}$$

Substitute the mixture survival probabilities to compute the conditional expression.

7. Poisson Random Variables

A **Poisson distribution** models counts of events occurring in a fixed interval.

If

$$X \sim \text{Poisson}(\lambda)$$

then

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

Mean and Variance

$$E[X] = \lambda$$

$$Var(X) = \lambda$$

Sum of Independent Poisson Variables

If

$$X \sim \text{Poisson}(\lambda_1), \quad Y \sim \text{Poisson}(\lambda_2)$$

and X and Y are independent, then

$$X + Y \sim \text{Poisson}(\lambda_1 + \lambda_2)$$

Conditional Distribution Given the Sum

If

$$X \sim \text{Poisson}(\lambda_1), \quad Y \sim \text{Poisson}(\lambda_2)$$

then conditioning on the total

$$N = X + Y$$

gives

$$X \mid (X + Y = n) \sim \text{Binomial}\left(n, \frac{\lambda_1}{\lambda_1 + \lambda_2}\right)$$

Thus

$$P(X = k \mid X + Y = n) = \binom{n}{k}\left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^k\left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{n-k}$$

8. Transformations of Random Variables

If a random variable Y is defined as a function of another variable

$$Y = g(X)$$

the distribution of Y can be obtained using either:

1. CDF method
2. Change of variables for pdf

CDF Method

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y)$$

Rewrite the inequality in terms of X, then evaluate the probability using the distribution of X.

PDF Transformation (One-to-One Case)

If $Y = g(X)$ is monotonic and differentiable,

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

where $x = g^{-1}(y)$.

Piecewise Transformations

If $g(X)$ is not one-to-one, each branch must be handled separately:

1. Solve $y = g(x)$ for all possible x
2. Add densities from each branch.

9. Approximations for Repeated Independent Trials

A model of repeated price movements or outcomes can often be described using **Bernoulli trials**.

Each trial has

- probability p of success
- probability $(1 - p)$ of failure.

Let

$$S_n = \text{number of successes in } n \text{ trials}$$

Then

$$S_n \sim \text{Binomial}(n, p)$$

Binomial Probability

$$P(S_n = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Normal Approximation

For large n, the binomial distribution can be approximated by

$$S_n \approx N(np, np(1-p))$$

This approximation is useful when n is very large.

10. Joint Distribution Transformations

If two independent continuous variables X and Y have densities

$$f_X(x), \quad f_Y(y)$$

then the joint density is

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

If a new variable is defined by

$$Z = \sqrt{X^2 + Y^2}$$

then determining the distribution of Z requires expressing the probability

$$P(Z \leq z)$$

in terms of the joint distribution of X and Y, typically involving integration over the region

$$x^2 + y^2 \leq z^2$$

This corresponds to integrating over a circular region in the (x, y)-plane.

11. Order Statistics and Maximum of Random Variables

Let

$$X_1, X_2, \ldots, X_n$$

be i.i.d. random variables with CDF $F(x)$.
Define

$$M_n = \max(X_1, \ldots, X_n)$$

Distribution of the Maximum

The CDF of the maximum is

$$P(M_n \leq x) = P(X_1 \leq x, \ldots, X_n \leq x)$$

Because the variables are independent,

$$F_{M_n}(x) = [F(x)]^n$$

The pdf is obtained by differentiating.

Joint Distribution of Successive Maxima

Consider

$$M_n = \max(X_1, \ldots, X_n)$$

and

$$M_{n+1} = \max(X_1, \ldots, X_n, X_{n+1})$$

Two cases arise:

1. $X_{n+1} \leq M_n$
2. $X_{n+1} > M_n$

The joint distribution is derived by analyzing these cases using independence and the distribution $F(x)$.

12. Key Techniques Used Throughout

The problems in the tutorial rely on repeatedly applying:

- Interpretation of probability models
- Writing events in terms of random variables
- Using pdf/CDF definitions
- Conditional probability and Bayes' theorem
- Expectation minimization
- Exponential memoryless property
- Conditioning on totals for Poisson variables
- Transformations of variables
- Approximation methods for large n
- Order statistics properties

Mastering these derivations and formulas allows one to systematically construct the probability expressions needed to solve each problem.

All of the above concepts form the mathematical foundation necessary to analyze the scenarios presented in Tutorial-2.